Interview preparation should be personal

PROBLEM & VALUE

AI Job Interview Research & Personalised Preparation Agent

The challenge

Job seekers spend time researching employers, interpreting job ads and translating their CV into convincing interview answers.

My solution

One form triggers company research, CV-to-role analysis and a structured seven-section preparation report.

The value

Relevant strengths, evidence gaps, tailored questions and STAR suggestions grounded in the applicant's own experience.

- **Presented by Martha Olukayode**
- **Project: AI Interview Preparation Agent**

From one submission to a ready-to-use report

No manual handoffs between research, analysis and delivery

01 INPUT

Company + job ad + PDF CV + consent

02 RESEARCH

Tavily + NewsAPI

03 RETRIEVE

Pinecone for SAP/Microsoft; fallback otherwise

04 ANALYSE

OpenAI GPT-4o-mini

05 DELIVER

HTML/PDF + Notion

Validation and API failure handling support the workflow; applicants must acknowledge data processing.

Evidence-aware by design

n8n orchestrates the process; retrieval is conditional on the hiring company

Research & RAG

Tavily: live company sources

NewsAPI: recent news

Pinecone: company-filtered SAP and Microsoft documents

Personalisation

OpenAI compares job requirements with CV evidence and produces seven sections. Siemens uses the live-research fallback.

Privacy & reliability

Mandatory consent acknowledgement; execution-saving settings configured; synthetic test CVs; API failure messaging; Notion delivery.

LIVE DEMO

Watch the agent work

Demonstrate a synthetic application from form submission to report

A | Submit

Open the production form. Show company, synthetic CV, job description and privacy consent.

Launch AI Interview Preparation Agent

B | Automate

Show n8n execution and the research → retrieval/fallback → OpenAI path.

C | Deliver

Open the seven-section HTML report; show Print / Save as PDF and the generated Notion page.

A focused MVP, ready for user validation

Why n8n now, and how I would test demand next

Why n8n?

Fast visual orchestration, API integration, branching and demo-ready delivery. LangGraph is a future option for complex stateful agents.

GTM: 3 experiments

1. Job seekers: usefulness feedback
2. Career coaches: pilot and repeat use
3. Onboarding: completion rate

What comes next

Strengthen source and STAR verification, complete resilience checks and validate usefulness with real users under appropriate privacy controls.

Thank you!

Questions?

AI Interview Preparation Agent

Autonomous research • CV-grounded insights • Interview readiness